



W3C VC WORKING GROUP · RECOGNIZED ENTITIES

Can a CE mark be a recognition chain?

Modelling EU product conformity on VC Recognized Entities v1.0 — what fits, what does not, and three questions for the group.

Spherity — verifiable credential infrastructure 2026-08 · v1.0

Why we brought this to the group

The exercise

The CE mark is a recognition hierarchy that already exists in law: named roles, statutory scope, real registries, and a verifier — the market surveillance authority — with actual powers. That makes it a useful stress test for a vocabulary that expresses recognition without defining a PKI.

We mapped it end to end. Most of it fits. Three things do not, and one of those is structural.

What we are asking for

We want the group's read on whether the places this model fails to reach are gaps in our reading of the spec or gaps in the spec itself. We are not asking for adoption, or for a review of our JSON.

Working Draft, 22 August 2026: "This is an experimental specification and is undergoing regular revisions. It is not fit for production deployment." We are reading it in that spirit.



CE marking in sixty seconds

WHO SAYS YOU ARE COMPETENT

Two separate acts

Each member state appoints a single national accreditation body. It assesses a conformity assessment body against the harmonised standards and any sectoral requirements.

Separately, a notifying authority designates that body and notifies the Commission and the other member states. The Commission assigns its identification number and publishes it in NANDO.

Accreditation is the preferred basis for designation, but designation is a separate decision.

WHO EXAMINES THE PRODUCT

The notified body

Where the directive requires it, the notified body examines the product type and issues an EU-type examination certificate.

Under the EMC Directive this is optional. A manufacturer may instead use internal production control and involve no third party at all.

WHO MAKES THE CLAIM

The manufacturer

The manufacturer draws up the EU Declaration of Conformity and, in drawing it up, assumes responsibility for compliance. The manufacturer affixes the CE marking.

Art. 30(1) of Reg. (EC) 765/2008 reserves affixing to the manufacturer or its authorised representative. The certificate feeds the declaration; it does not stand in for it.

Reg. 765/2008 Art. 30(1) · 2014/30/EU Art. 7(2), 15(4)

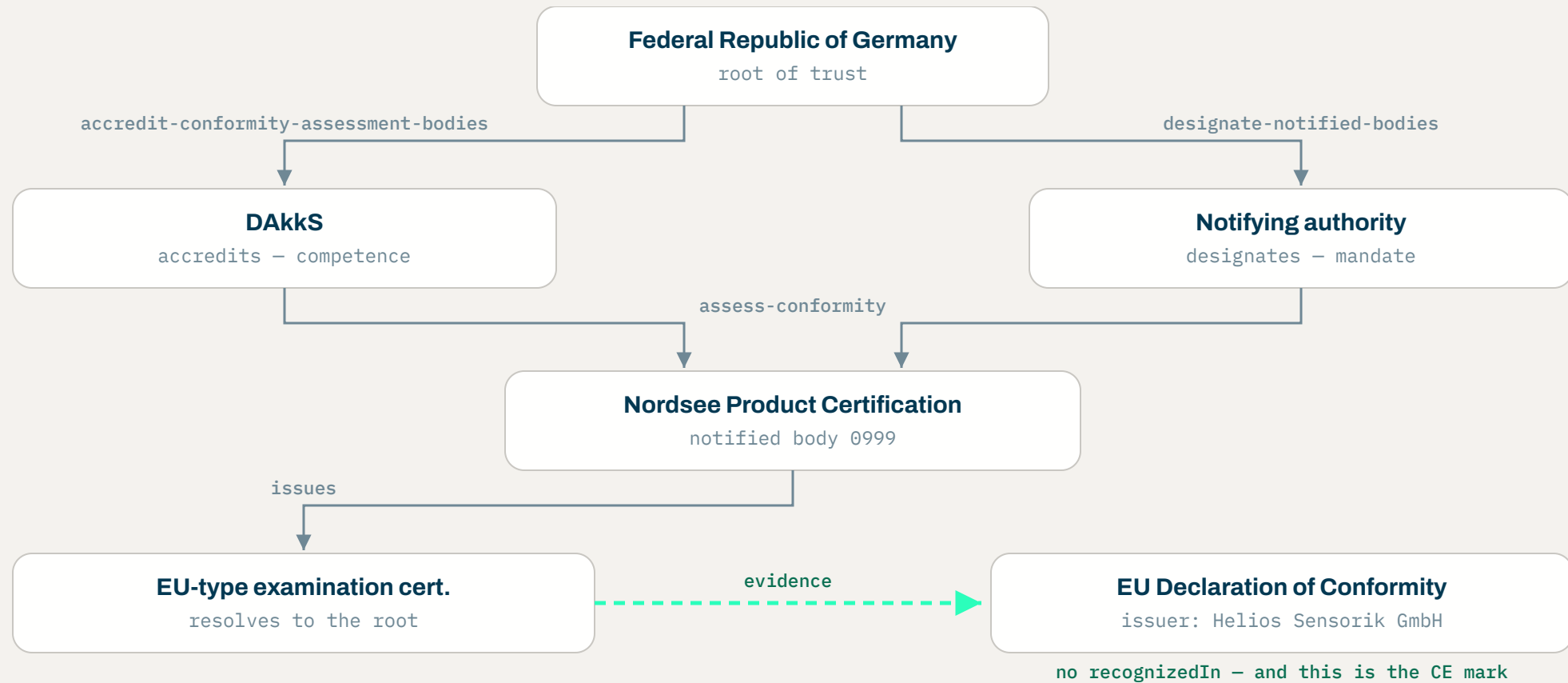


The mapping

RECOGNIZED ENTITIES	CE MARKING ROLE	EXAMPLE (DID:WEB:)
Root of trust	EU member state	gov.de
Tier 1 · accredit-conformity-assessment-bodies	National accreditation body, Reg. (EC) 765/2008	dakks.de
Tier 1 · designate-notified-bodies	Notifying authority — a ministry or its delegate	notifying-authority.bund.de
Tier 2 · assess-conformity	Notified body — notified under 2014/30/EU Art. 20	nordsee-certification.eu
(no recognized entity)	Manufacturer: draws up the declaration, affixes the mark	helios-sensorik.example

Accreditation and designation are separate acts by separate authorities, though a member state may have the accreditation body carry out the assessment for the notifying authority (Art. 21(2)). The last row issues the artifact that matters and holds no recognition at all.

The chain, drawn honestly



Tier 1 — two recognizers, two credentials

DOC A1 — MEMBER STATE → ACCREDITATION BODY

```
{
  "id": "https://gov.de/recognized/accred-bodies",
  "type": ["VerifiableCredential", "RecognizedEntityCredential"],
  "issuer": { "id": "did:web:gov.de",
    "type": "RecognizedIssuer",
    "name": "Federal Republic of Germany" },
  "credentialSubject": [{
    "id": "did:web:dakks.de",
    "type": "RecognizedEntity",
    "name": "Deutsche Akkreditierungsstelle GmbH",
    "recognizedTo": {
      "type": "RecognizedAction",
      "action": "accredit-conformity-assessment-bodies",
      "recognizedBy": "did:web:gov.de",
      "validUntil": "2030-12-31T23:59:59Z"
    }
  }], "proof": { ... }
}
```

DOC A2 — MEMBER STATE → NOTIFYING AUTHORITY

```
{
  "id": "https://gov.de/recognized/notifying-authorities",
  "type": ["VerifiableCredential", "RecognizedEntityCredential"],
  "issuer": { "id": "did:web:gov.de",
    "type": "RecognizedIssuer",
    "name": "Federal Republic of Germany" },
  "credentialSubject": [{
    "id": "did:web:notifying-authority.bund.de",
    "type": "RecognizedEntity",
    "name": "Notifying authority (2014/30/EU Art. 21)",
    "recognizedTo": {
      "type": "RecognizedAction",
      "action": "designate-notified-bodies",
      "recognizedBy": "did:web:gov.de"
    }
  }], "proof": { ... }
}
```

Both actions are free-form strings with no registry. Nothing in the credentials says the two are related, or that a notified body needs both.

Tier 2 — one body, two parents

DOC B1 — ACCREDITATION

```
{
  "id": "https://dakks.de/recognized/cabs",
  "issuer": { "id": "did:web:dakks.de",
    "type": "RecognizedIssuer",
    "recognizedIn": {
      "id": "https://gov.de/recognized/accred-bodies",
      "type": "RecognizedEntityCredential" }
    },
  "credentialSubject": [{
    "id": "did:web:nordsee-certification.eu",
    "type": "RecognizedEntity",
    "recognizedTo": {
      "action": "assess-conformity",
      "recognizedBy": "did:web:dakks.de",
      "outputValidation": { "type": "JsonSchema", ... },
      "validUntil": "2029-03-14T23:59:59Z"
    }
  ]
}
```

DOC B2 — NOTIFICATION

```
{
  "id": "https://...bund.de/recognized/notified-bodies",
  "issuer": { "id": "did:web:notifying-authority.bund.de",
    "type": "RecognizedIssuer",
    "recognizedIn": {
      "id": "https://gov.de/recognized/notifying-authorities",
      "type": "RecognizedEntityCredential" }
    },
  "credentialSubject": [{
    "id": "did:web:nordsee-certification.eu",
    "type": "RecognizedEntity",
    "description": "Notified body 0999",
    "recognizedTo": {
      "action": "assess-conformity",
      "recognizedBy": "did:web:notifying-authority.bund.de",
      "validUntil": "2029-03-14T23:59:59Z"
    }
  ]
}
```

Both are valid chains to a trusted root. A verifier that walks only B1 has confirmed competence and missed the mandate; walking only B2 inverts that.

The chain anchors the certificate, not the mark

DOC C – THE NOTIFIED BODY'S CERTIFICATE

```
{
  "type": ["VerifiableCredential", "EUTypeExaminationCertificate"],
  "id": "https://nordsee-certification.eu/c/0999-2026-04471",
  "issuer": {
    "id": "did:web:nordsee-certification.eu",
    "type": "RecognizedIssuer",
    "recognizedIn": {
      "id": "https://...bund.de/recognized/notified-bodies",
      "type": "RecognizedEntityCredential" }
  },
  "credentialSubject": {
    "type": "ProductType",
    "name": "Industrial Temp. Sensor HS-TS-200",
    "directive": "2014/30/EU (EMC), Annex III Part A",
    "module": "Module B – EU-type examination"
  }
}
```

DOC D – THE DECLARATION, AND THE CE MARK

```
{
  "type": ["VerifiableCredential", "EUDeclarationOfConformity"],
  "issuer": {
    "id": "did:web:helios-sensorik.example",
    "name": "Helios Sensorik GmbH"
  },
  "credentialSubject": {
    "type": "Product",
    "name": "Industrial Temp. Sensor HS-TS-200",
    "ceMarking": true,
    "directives": ["2014/30/EU"]
  },
  "evidence": [{
    "type": "EUTypeExaminationCertificate",
    "id": "https://nordsee-certification.eu/c/0999-2026-04471"
  }]
}
```

Doc C resolves to a root. Doc D has no recognizedIn, and Doc D is the artifact a market surveillance authority acts on. The spec defines recognizedIn as a single object, so Doc C can name only one of its two parents.



How a verifier walks it

CREDENTIAL-BASED DISCOVERY, §4.1

- 1 Start at Doc C. If its issuer is already in the verifier's recognized set, stop.
- 2 Otherwise read issuer.recognizedIn, fetch that list, verify its proof, confirm the list is currently valid, and confirm Doc C's issuer appears as a recognized entity in its credentialSubject.
- 3 Append to the chain and repeat one tier up, until an issuer is recognized or maxDepth is exceeded.
- 4 §4.1 returns the chain and nothing else. Matching the action to the credential type, and checking outputValidation, is §4.2.

And then it stops

Doc D's issuer has no recognizedIn, so there is nothing to follow. A verifier can check that Doc D cites Doc C as evidence and that Doc C's chain resolves. But the step from “this body was recognized” to “this manufacturer may therefore declare” is not in the credential graph. It is in the directive.

A holder may staple the chain so the verifier fetches nothing, provided the verifier is satisfied with its freshness. Stapling cannot reach Doc D, which was never in the chain.



Where the mapping breaks

The body issues no mark

Reg. (EC) 765/2008 Art. 30(1): the CE marking is affixed only by the manufacturer or its authorised representative. Decision 768/2008 Art. R10: by drawing up the declaration, the manufacturer assumes responsibility for compliance. No module in Annex II reassigns either act.

Accreditation is not notification

One body assesses competence under Reg. (EC) 765/2008; a notifying authority designates, and the Commission assigns the identification number (Art. 29(1)). Art. 21(2) lets a member state hand the assessment back to the accreditation body, so the shape is a graph, not a chain, with an edge that varies by member state.

Most products skip the body

Under EMC Art. 14 a manufacturer may choose internal production control alone (Annex II, Module A). The mark then rests on its own technical file, with no recognition chain to walk at all. For most apparatus this is the ordinary route.

Three questions for the group

An issuer authorised by statute

Doc D's issuer holds no recognition. Its right to issue comes from the directive, conditioned on holding Doc C. Is that a RecognizedEntity with an implicit root, an evidence pattern the spec should describe, or deliberately out of scope?

Two recognizers, one action

recognizedTo may occur more than once, but recognizedIn is defined as a single object. Competence and mandate come from different parents. May recognizedIn be plural, and if it may, how does a verifier learn that both links are required rather than either one?

Revocation and withdrawal

Notification can be restricted, suspended or withdrawn (Art. 30), and the member state must then ensure the body's files are taken over. The spec defines no revocation of its own, and §4.1 asks only whether a list is currently valid, never whether an action was valid on the examination date. What is the intended pattern?

Tell us where this mapping is wrong

`info@spherity.com`

`spherity.com`

`w3.org/TR/vc-recognized-entities-1.0/`

